Analysis-0 2026 Spring

Homework 7

程昊一

*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

$\boxed{\textbf{Problem 1} \text{ (4-2).}}$ Let f be a continuous mapping from a metric space X into a metric space Y. Prove that for every set $E \subset X$,

$$f(\overline{E}) \subset \overline{f(E)},$$

where $\overline{E}$ denotes the closure of E. Give an example showing that $f(\overline{E})$ can be a proper subset of $\overline{f(E)}$.

Proof. Take $p \in \overline{E}$. We will show that $f(p) \in \overline{f(E)}$, i.e. for any $\varepsilon > 0$, there exists $q \in E$ such that $d(f(p), f(q)) < \varepsilon$.

Since f is continuous at p, there exists $\delta > 0$ such that for any $x \in X$ with $d(p, x) < \delta$, we have $d(f(p), f(x)) < \varepsilon$. Since $p \in \overline{E}$, there exists $q \in E$ such that $d(p, q) < \delta$. Thus, we have $d(f(p), f(q)) < \delta$, which completes the proof of $f(\overline{E}) \subset \overline{f(E)}$.

For the second part, let $X = Y = \mathbb{R}$ with the metric $d(x, y) = |x - y|$, and $f(x) = \arctan x$. Then, for $E = \mathbb{R}$, we have $f(E) = f(\overline{E}) = (-\pi/2, \pi/2)$ but $\overline{f(E)} = [-\pi/2, \pi/2]$, which shows that $f(\overline{E})$ is a proper subset of $\overline{f(E)}$. $\square$

$\boxed{\textbf{Problem 2} \text{ (4-4).}}$ Let f and g be continuous mappings from a metric space X into a metric space Y, and let E be a dense subset of X.

Prove that $f(E)$ is dense in $f(X)$. If $g(p) = f(p)$ for every $p \in E$, prove that $g(p) = f(p)$ for every $p \in X$. (In other words, a continuous mapping is determined by its values on a dense subset of its domain.)

Proof. By the previous problem, we have $f(\overline{E}) \subset \overline{f(E)}$. Since E is dense in X, we have $\overline{E} = X$, which implies that $f(X) \subset \overline{f(E)}$. Thus, $f(E)$ is dense in $f(X)$.

For the second part, let

$$h: \ X \to \mathbb{R}, \ x \to d(f(x), g(x)).$$

It's not hard to verify that h is continuous on X and $h(p) = 0$ for every $p \in E$. Since E is dense in X, we have $\{0\}$ is dense in $h(X)$. Since $h(X)$ is a subset of $\mathbb{R}$, we have $h(X) = \{0\}$, which implies that $f(p) = g(p)$ for every $p \in X$. $\square$

Problem 3 (4-6). If f is defined on E, then the graph of f is the set of points $(x, f(x))$ with $x \in E$. In particular, if E is a subset of the real numbers and f is real-valued, then the graph of f is a subset of the plane.

Assume E is compact. Prove that f is continuous on E if and only if its graph is compact.

Proof. Denote $g(x) = (x, f(x))$ as a function from E to $\mathbb{R}^2$, and define $S = g(E)$. Then g is continuous if f is continuous. Assume that $E \times f(E)$ is equipped with the product topology.

(1) " $\Longrightarrow$ ". For any open cover $\{G_\alpha\}$ of S, we have $\tilde{G}_\alpha := g^{-1}(G_\alpha \cap S)$ is an open cover of E. Since E is compact, there exists a finite subcover $\{\tilde{G}_{\alpha_i}\}$ of E. Then, $\{G_{\alpha_i}\}$ is a finite subcover of S, which implies that S is compact.

(2) " $\Longleftarrow$ ". For any $x \in E$ and $\varepsilon > 0$, we will prove that there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for any $y \in N_\delta(x)$.

Denote

$$G = N_\varepsilon(x) \times N_\varepsilon(f(x)),\ G_n = \big(E - \overline{N}_{\varepsilon/n}(x)\big) \times f(E),$$

where $\overline{N}$ denote the closed neighborhood. Then, by the definition of product topology, G and G_n is open in $E \times f(E)$.

Since for every $y \neq x$, we have $(y, f(y)) \in G_n$ for sufficiently large n, and $(x, f(x)) \in G$, therefore $\{G\} \cup \{G_n :\ n \in \mathbb{N}\}$ is an open cover of S. Since S is compact, there exists a finite subcover $\{G\} \cup \{G_{n_i} :\ i = 1, 2, \cdots, k\}$ of S. Let $N = \max\{n_i :\ i = 1, 2, \cdots, k\}$. Then, for any $y \in E$ with $|x - y| < \varepsilon/N$, we have $(y, f(y)) \in G$, which implies that $|f(x) - f(y)| < \varepsilon$.

Take $\delta = \varepsilon/N$, we have $|f(x) - f(y)| < \varepsilon$ for any $y \in N_\delta(x)$, which completes the proof. □

Problem 4 (4-8). Let f be a uniformly continuous real-valued function on a bounded set $E \subset \mathbb{R}^1$. Prove that f is bounded on E.

If the boundedness assumption on E is removed, show that this conclusion need not hold.

Proof. Since $f(E)$ is not bounded, there exists a sequence $\{x_n\} \in E$ such thath $|f(x_{n+1})| > 1 + |f(x_n)|$ for all $n \geq 1$. Since E is bounded, there exists a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that $x_{n_k} \to x_0$ for some $x_0 \in \mathbb{R}$. Since f is uniformly continuous on E, then for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x, y \in E$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$. Take k_0 such that $|x_{n_k} - x_0| < \delta/2$ for any $k \geq k_0$. Then, for any $k, l \geq k_0$, we have

$$|f(x_{n_k}) - f(x_{n_l})| < \varepsilon,$$

which contradicts the fact that $|f(x_{n+m})| > m + |f(x_n)|$ for all $m, n \geq 1$.

For the second part, let $E = \mathbb{R}$ and $f(x) = x$. Then, f is uniformly continuous on E but $f(E)$ is not bounded. □

Problem 5 (4-9). Prove that the requirement in the definition of uniform continuity can be expressed in terms of diameters as follows: for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every $E \subset X$ with $\operatorname{diam}(E) < \delta$, one has $\operatorname{diam}(f(E)) < \varepsilon$.

Proof. First, we will show that the original definition of uniform continuity implies the diameter version. For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x, y \in X$ with $d(x, y) < \delta$, we have $d(f(x), f(y)) < \varepsilon$. For any $E \subset X$ with $\operatorname{diam}(E) < \delta$, we have $d(x, y) < \delta$ for any $x, y \in E$. Thus, we have $d(f(x), f(y)) < \varepsilon$ for any $x, y \in E$, which implies that $\operatorname{diam}(f(E)) < \varepsilon$.

Next, we will show that the diameter version implies the original definition of uniform continuity. For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $E \subset X$ with $\operatorname{diam}(E) < \delta$, we have $\operatorname{diam}(f(E)) < \varepsilon$. For any $x, y \in X$ with $d(x, y) < \delta$, we have $\operatorname{diam}(\{x, y\}) = d(x, y) < \delta$. Thus, we have $\operatorname{diam}(f(\{x, y\})) < \varepsilon$, which implies that $d(f(x), f(y)) < \varepsilon$.

This completes the proof. □

Problem 6 (4-12). The composition of uniformly continuous functions is uniformly continuous. State this assertion precisely, and prove it.

Proof. Let X, Y, Z be metric spaces, and $f : X \to Y$, $g : Y \to Z$ be uniformly continuous functions. We will show that the composition $g \circ f : X \to Z$ is uniformly continuous.

For any $\varepsilon > 0$, since g is uniformly continuous on Y, there exists $\delta_1 > 0$ such that for any $y_1, y_2 \in Y$ with $d(y_1, y_2) < \delta_1$, we have $d(g(y_1), g(y_2)) < \varepsilon$. Since f is uniformly continuous on X, there exists $\delta_2 > 0$ such that for any $x_1, x_2 \in X$ with $d(x_1, x_2) < \delta_2$, we have $d(f(x_1), f(x_2)) < \delta_1$. Thus, for any $x_1, x_2 \in X$ with $d(x_1, x_2) < \delta_2$, we have

$$d(g(f(x_1)), g(f(x_2))) < \varepsilon,$$

which implies that $g \circ f : X \to Z$ is uniformly continuous.

This completes the proof. □